ARCHITECTURE & INFRASTRUCTURE OVERVIEW

The AI Software Factory

Autonomous AI agents converting approved specifications into pull requests under strict, human-approved merges and machine-enforced governance.

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CORE PARADIGM

Enforced Execution Over Written Documentation

> **“A standard counts only when it is enforced and running, not when it is written down.”**
>
> Every requirement must pass automated, zero-trust quality gates prior to human approval and deployment.

GOVERNANCE

Engineering Excellence Playbook

Vendored Rulebooks

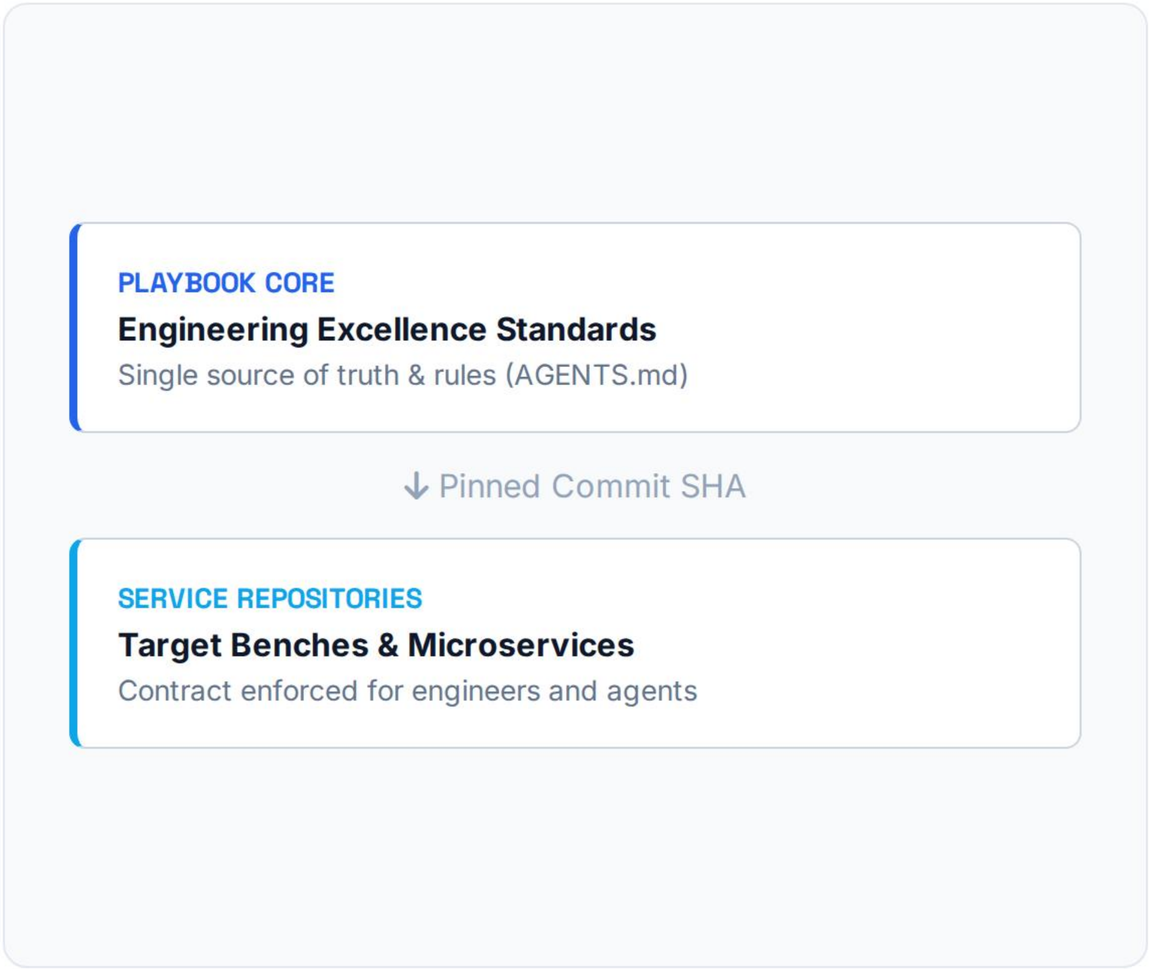
Vendors `AGENTS.md` at pinned commit SHAs directly into service repositories.

Schema Contracts

Versioned JSON Schemas backed by automated test vectors for deterministic behavior.

Evidence-Based Progression

Program Increments grading factory score from a 61 baseline toward 90+ maturity.



MODULAR TOPOLOGY

Specialized Factory Benches

Central CI

Reusable quality gate executed on every commit SHA.

Builder

Automated PR generation directly from approved specs.

Reviewer

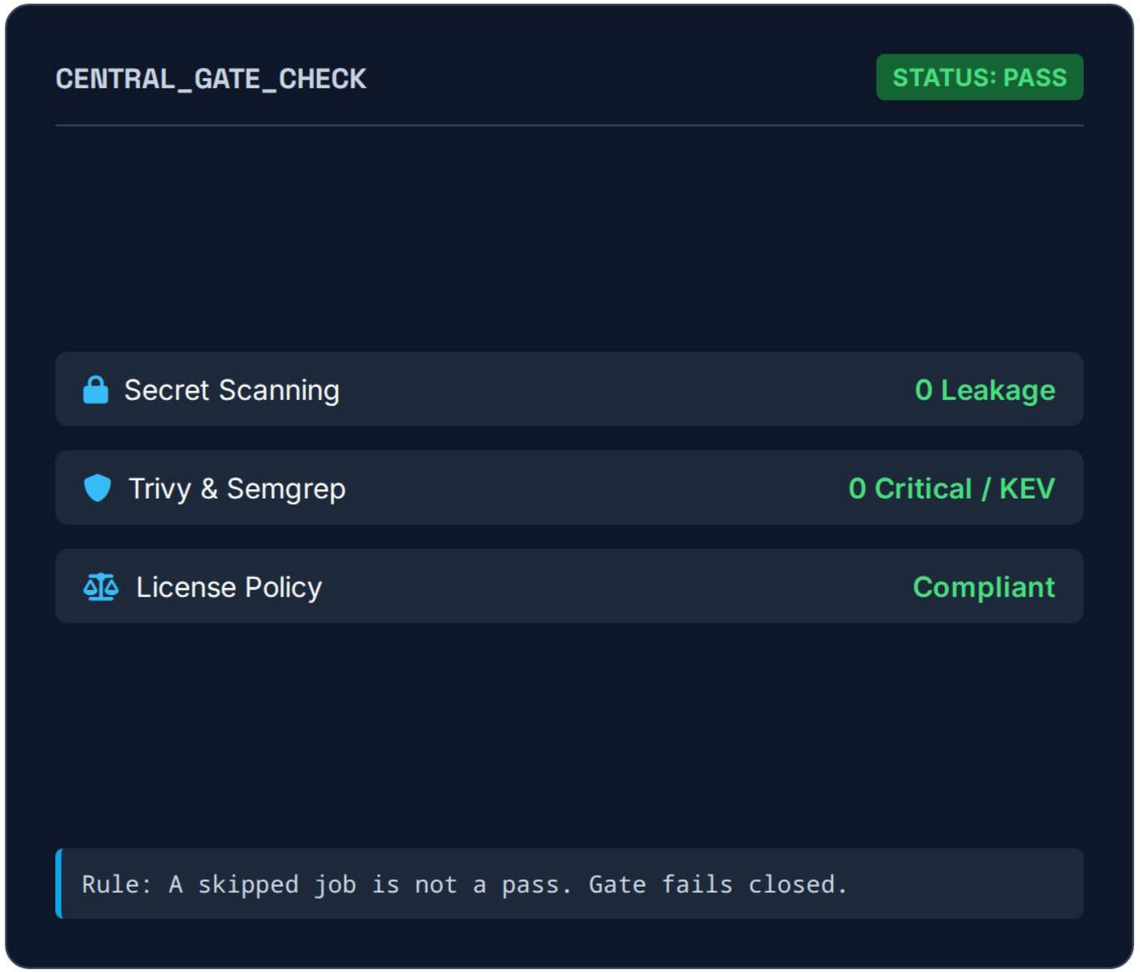
Multi-lens analysis for security and architecture.

Compliance

Verifies evidence & issues single-use tokens (In Progress).

SECURITY MODEL

Zero-Trust Quality Gates



Multi-Scanner Enforcement

Integrated secret scanning, Trivy, Semgrep, and automated license validation on every commit.

Fail-Closed Policy

Any CRITICAL flaw or CISA KEV vulnerability immediately halts the pipeline.

Strict Verification Rule

Fixture projects prove defect detection; a skipped job is never counted as a pass.

ISOLATION & PACKAGING

Hermetic Container Builds

Unprivileged Runtime

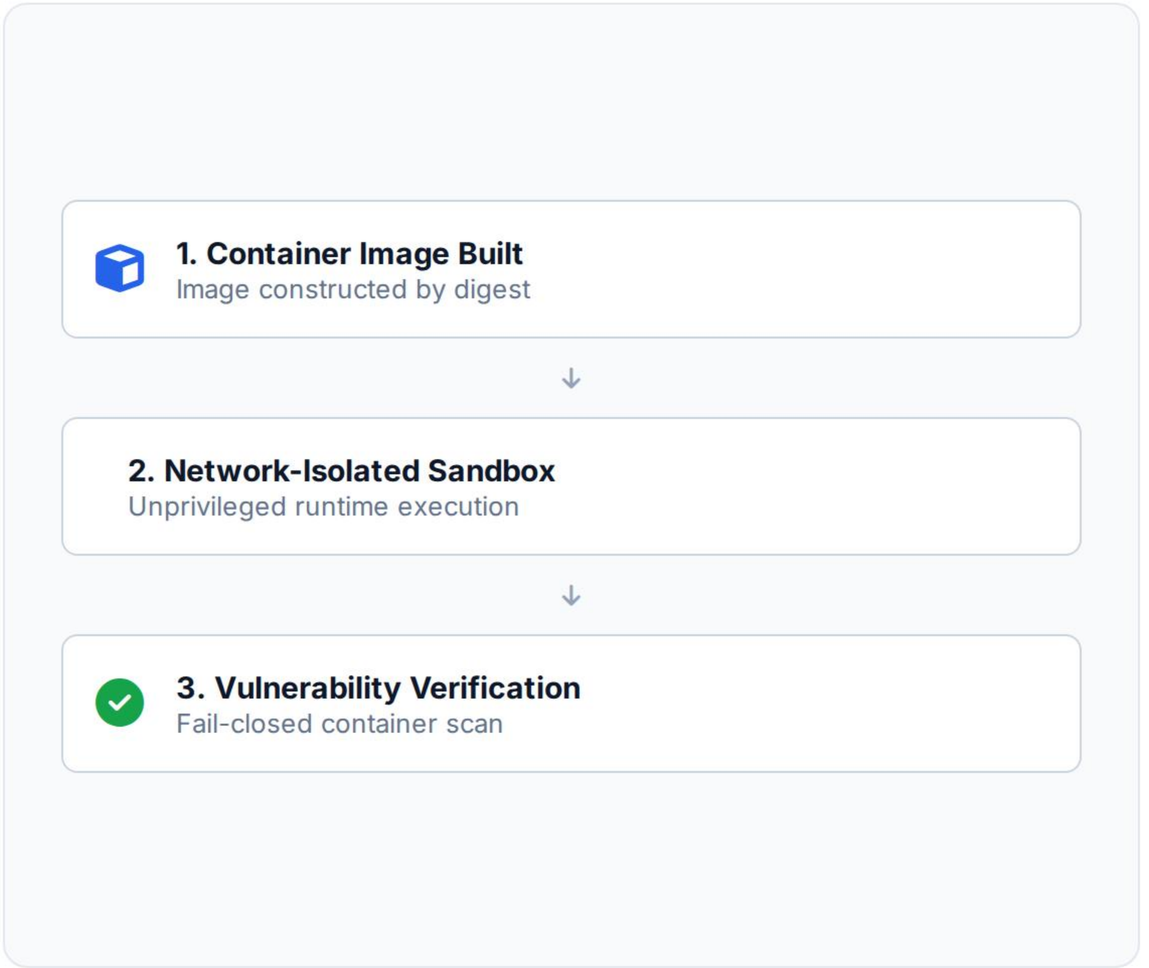
Tests execute inside non-root containers without external network access.

Zero Exception Remediation

Vulnerabilities are resolved at the upstream source rather than bypassed with exceptions.

Immutable Dependency Pinning

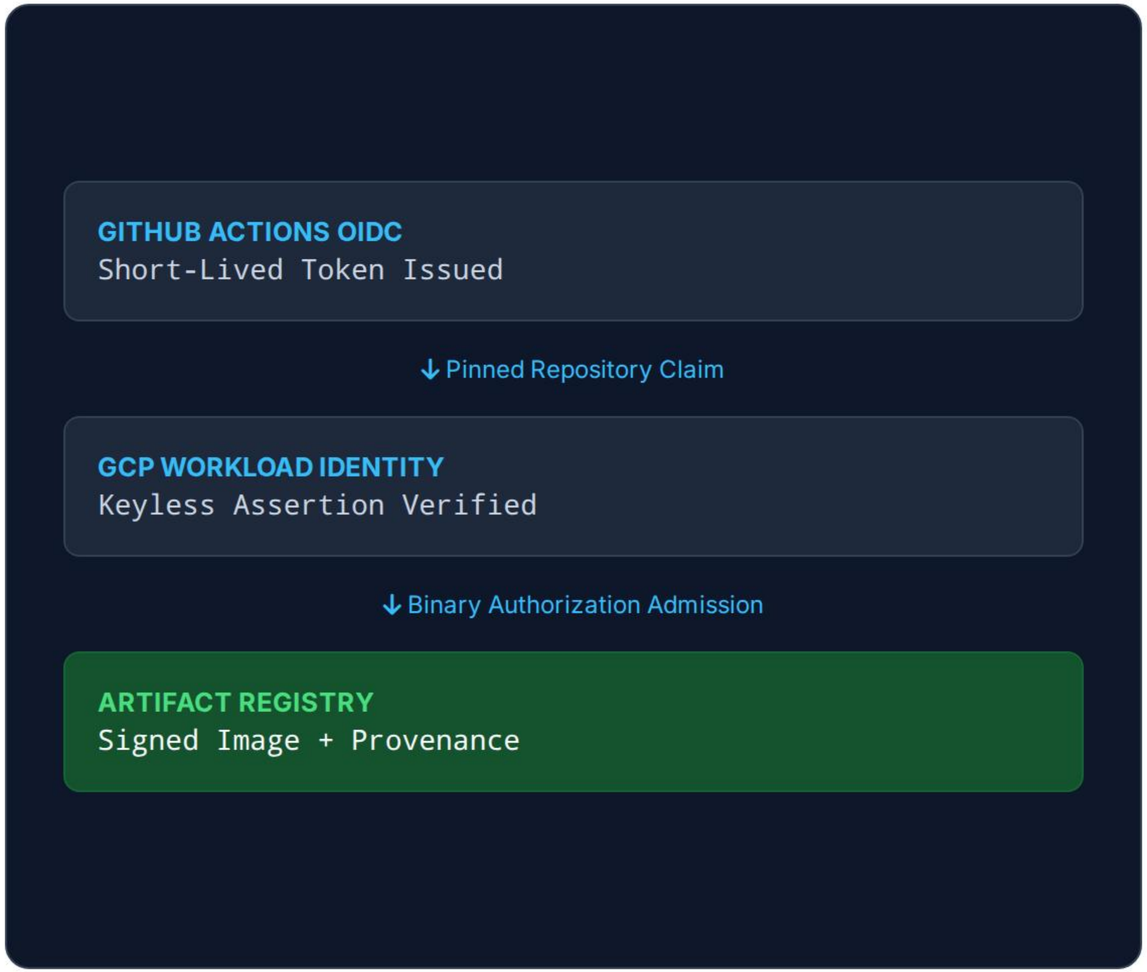
Strict SHA commit pinning for actions, container digests, and tool checksums.



CONTINUOUS DELIVERY

IN PROGRESS HORIZON

Planned Keyless Supply Chain Architecture



Workload Identity Federation

Eliminates persistent service account keys via strict GitHub assertion pins.

Provenance & Attestation (Planned)

Publishing SLSA provenance, SBOM, and keyless signatures on merge to main.

Binary Authorization (Planned)

Enforcing deployment guardrails to admit only verified images.

ARCHITECTURE PATTERN

IN PROGRESS HORIZON

Event-Driven Loop Architecture Design

STAGE 01

Outbox Pattern

Guaranteed transaction logging for handoffs.

STAGE 02

Agent Inbox

Idempotent processing across benches.

STAGE 03

Dead-Letter Queues

Fault isolation for unhandled events.

GOVERNANCE TRANSITION

Autonomy Earned Through Evidence

61 → 90+

Program Increment Quality Score Target

Human approval remains required at merge gates until automated measurements prove stability and quality without intervention.

ENGINEERING EXCELLENCE PLAYBOOK

Building Software Factories that Run on Proof

github.com/Drecoder/Engineering_Excellence_Playbook